

Homework 2

1)

Let $e \in \text{Exp}$. We prove by structural induction on e that:

$$\forall \sigma \in \text{Store}. \exists \sigma' \in \text{Store}. \exists e' \in \text{Exp}. \langle \sigma, e \rangle \longrightarrow^* \langle \sigma', e' \rangle \wedge \langle \sigma', e' \rangle \not\longrightarrow.$$

Case $e = x$

- $x \in \sigma$: By VAR, $\langle \sigma, x \rangle \longrightarrow \langle \sigma, n \rangle$. By REFL and TRANS, $\langle \sigma, x \rangle \longrightarrow^* \langle \sigma, n \rangle$. Because $\langle \sigma, n \rangle \not\longrightarrow$, done.
- $x \notin \sigma$: By REFL, $\langle \sigma, x \rangle \longrightarrow^* \langle \sigma, x \rangle$. Because $\langle \sigma, x \rangle \not\longrightarrow$ if $x \notin \sigma$, done.

Case $e = n$

By REFL, $\langle \sigma, n \rangle \longrightarrow^* \langle \sigma, n \rangle$. Because $\langle \sigma, n \rangle \not\longrightarrow$, done.

Case $e = e_1 + e_2$ (same for $e_1 * e_2$)

By IH1, $\langle \sigma, e_1 \rangle \longrightarrow^* \langle \sigma', e'_1 \rangle \wedge \langle \sigma', e'_1 \rangle \not\longrightarrow$.

- $e'_1 = n$: By Lemma 1 (proven during lecture), $\langle \sigma, e_1 + e_2 \rangle \longrightarrow^* \langle \sigma', n + e_2 \rangle$.
By IH2, $\langle \sigma', e_2 \rangle \longrightarrow^* \langle \sigma'', e'_2 \rangle \wedge \langle \sigma'', e'_2 \rangle \not\longrightarrow$.
 - $e'_2 = m$: By Lemma 1, $\langle \sigma', n + e_2 \rangle \longrightarrow^* \langle \sigma'', n + m \rangle$. By transitivity, $\langle \sigma, e_1 + e_2 \rangle \longrightarrow^* \langle \sigma'', p \rangle$, where $p = n + m$. Because $\langle \sigma'', p \rangle \not\longrightarrow$, done.
 - $e'_2 \neq m$: By Lemma 1, $\langle \sigma', n + e_2 \rangle \longrightarrow^* \langle \sigma'', n + e'_2 \rangle$. By transitivity, $\langle \sigma, e_1 + e_2 \rangle \longrightarrow^* \langle \sigma'', n + e'_2 \rangle$. Because $\langle \sigma'', e'_2 \rangle \not\longrightarrow$ and $e'_2 \neq m$, $\langle \sigma'', n + e'_2 \rangle \not\longrightarrow$, done.
- $e'_1 \neq n$: By Lemma 1, $\langle \sigma, e_1 + e_2 \rangle \longrightarrow^* \langle \sigma', e'_1 + e_2 \rangle$. Because $\langle \sigma', e'_1 \rangle \not\longrightarrow$ and $e'_1 \neq n$, $\langle \sigma', e'_1 + e_2 \rangle \not\longrightarrow$, done.

Case $e = x := e_1; e_2$

By IH1, $\langle \sigma, e_1 \rangle \longrightarrow^* \langle \sigma', e'_1 \rangle \wedge \langle \sigma', e'_1 \rangle \not\longrightarrow$.

- $e'_1 = n$: By Lemma 1, $\langle \sigma, x := e_1; e_2 \rangle \longrightarrow^* \langle \sigma', x := n; e_2 \rangle$. By ASSIGN, REFL, TRANS and transitivity, $\langle \sigma, x := e_1; e_2 \rangle \longrightarrow^* \langle \sigma'', e_2 \rangle$. By IH2 and transitivity, done.
- $e'_1 \neq n$: By Lemma 1, $\langle \sigma, x := e_1; e_2 \rangle \longrightarrow^* \langle \sigma', x := e'_1; e_2 \rangle$. Because $e'_1 \neq n$ and $\langle \sigma', e'_1 \rangle \not\longrightarrow$, $\langle \sigma', x := e'_1; e_2 \rangle \not\longrightarrow$. Done.

2)

- a. $e = x := 1; 0$
- b. $e_1 = x := 1; 0 \quad e_2 = x := 2; 0$
- c. Ok.

3)

Let $\sigma, \sigma' \in Store$, $c, c' \in Com$. We prove by induction on the derivation $\langle \sigma, c \rangle \longrightarrow^* \langle \sigma', c' \rangle$ that

$$\forall \sigma'', c''. \quad \underbrace{\langle \sigma, c \rangle \longrightarrow^* \langle \sigma', c' \rangle}_{(H1)} \wedge \underbrace{\langle \sigma', c' \rangle \longrightarrow^* \langle \sigma'', c'' \rangle}_{(H2)} \implies \langle \sigma, c \rangle \longrightarrow^* \langle \sigma'', c'' \rangle.$$

Case Refl

By H1, H2, $\sigma = \sigma' \wedge \sigma' = \sigma'' \wedge c = c' \wedge c' = c''$. By transitivity of $=$, $\sigma = \sigma'' \wedge c = c''$. By REFL, $\langle \sigma, c \rangle \longrightarrow^* \langle \sigma, c \rangle$.

Case Step

By STEP and H1, $\exists \sigma^*, c^*. \langle \sigma, c \rangle \longrightarrow \langle \sigma^*, c^* \rangle \wedge \langle \sigma^*, c^* \rangle \longrightarrow^* \langle \sigma', c' \rangle$.

By IH and H2, $\langle \sigma^*, c^* \rangle \longrightarrow^* \langle \sigma'', c'' \rangle$.

By STEP, $\langle \sigma, c \rangle \longrightarrow^* \langle \sigma'', c'' \rangle$.

4)

$$\frac{\langle \sigma, e_1 \rangle \longrightarrow \langle \sigma', e'_1 \rangle}{\langle \sigma, \text{try } e_1 \text{ with } E \text{ do } e_2 \rangle \longrightarrow \langle \sigma', \text{try } e'_1 \text{ with } E \text{ do } e_2 \rangle}$$

$$\overline{\langle \sigma, \text{try } E \text{ with } E \text{ do } e_2 \rangle \longrightarrow \langle \sigma, e_2 \rangle}$$

$$\overline{\langle \sigma, \text{try } n \text{ with } E \text{ do } e_2 \rangle \longrightarrow \langle \sigma, n \rangle}$$

$$\frac{E' \neq E}{\langle \sigma, \text{try } E' \text{ with } E \text{ do } e_2 \rangle \longrightarrow \langle \sigma, E' \rangle}$$

$$\left. \begin{array}{l} \overline{\langle \sigma, \text{throw } E \rangle \longrightarrow \langle \sigma, E \rangle} \\ \frac{\langle \sigma, E + e_2 \rangle \longrightarrow \langle \sigma, E \rangle}{\langle \sigma, n + E \rangle \longrightarrow \langle \sigma, E \rangle} \end{array} \right\} \text{ same for } e_1 * e_2$$

$$\overline{\langle \sigma, x := E; e_2 \rangle \longrightarrow \langle \sigma, E \rangle}$$

Generative AI Disclosure

Claude Sonnet LLM was used to translate handwriting to LaTeX. In addition, Gemini Pro was used to understand what the inductive hypothesis of Exercise 3 should be in the Step case, as I didn't fully understand induction on derivations. The LLMs helped me not waste time by rewriting everything and aided my understanding of some topics.

Debriefing

- 4h
- moderate-difficult
- only me
- 90%